INCH-POUND
MIL-DTL-27467B
9 September 2005
SUPERSEDING
MIL-H-27467A (USAF)
1 March 1963

DETAIL SPECIFICATION

HEADSET, ELECTRICAL H-154A/AIC

Inactive for new design after 20 August 1996.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of headset, designated Headset, Electrical H-154A/AIC.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-25670/2	Earphone Element, High- and Low-Altitude, H-143/AIC, and Low-Altitude, Water-Immersible, H-143A/AIC
MIL-DTL-22442	Cable Assemblies, Aircraft Audio, General Specification For
MIL-DTL-22442/33	Cable Assemblies, Aircraft Audio, CX-4707C/AIC
MIL-DTL-22442/37	Cable Assembly, Aircraft Audio, CX-4708A/AIC
MIL-DTL-9177/2	Connector, Audio, Airborne, Plug, Miniature, 4 Contact

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-202	Test Method Standard, Electronic and Electrical Component Parts
MIL-STD-810	Environmental Engineering Considerations and Laboratory Tests
MIL-STD-1285	Marking of Electrical and Electronic Parts

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or http://assist.daps.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

Comments, suggestions, or questions on this document should be addressed to Defense Supply Center, Columbus, ATTN: DSCC-VAI, P.O. Box 3990, Columbus, OH 43218-3990, or email to Sound@dscc.dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil.

AMSC N/A FSC 5965

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T12:06Z
Check the source to verify that this is the current version before use.

DRAWINGS

Air Force

7136032	Headset Kit, MK-634/AIC
7136036	Headset, Electrical H-154A/AIC – Assembly of
7136037	Shell, Earphone – Headset
7136038	Cushion, Earphone – Headset
7136039	Pad – Earphone

(Copies of these drawings required by contractors in connection with specific procurement functions are available from the Defense Supply Center Columbus, ATTN: DSCC-VTRD, P.O. Box 3990, Columbus, Ohio 43218-3990.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

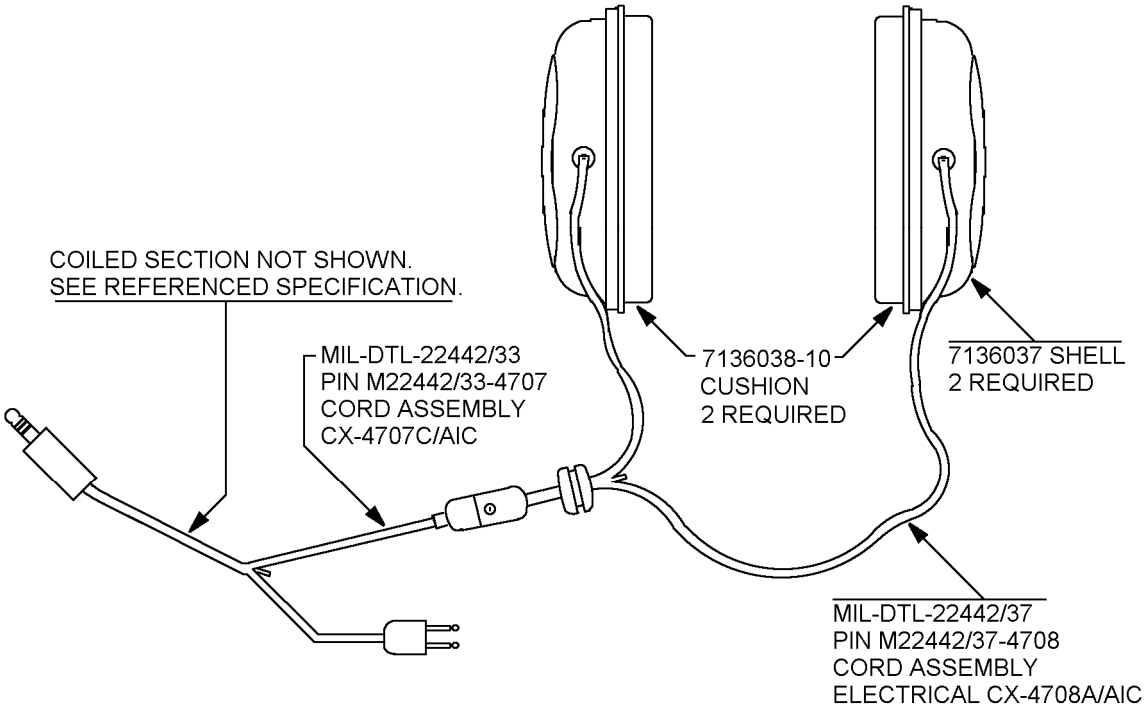


FIGURE 1. H-154A/AIC.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Materials. Materials shall be as specified herein. However, when a definite material is not specified, a material shall be used which will enable the product to meet the performance requirements of this specification. Acceptance or approval of any constituent material shall not be construed as a guaranty of the acceptance of the finished product.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T12:06Z
Check the source to verify that this is the current version before use.

3.2.1 Pure tin. The use of pure tin, as an underplating or final finish, is prohibited both internally and externally. Tin content of headset components and solder shall not exceed 97 percent. Tin shall be alloyed with a minimum of 3 percent lead (see 6.3).

3.2.2 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3 Construction.

3.3.1 General design. The headset shall be designed to be inherently stable in mechanical construction (see figure 1), electrical characteristics, and acoustical performance and shall be suitable for extended military aircraft use. The detailed mechanical and electrical design of this headset shall be accomplished by the contractor subject to the requirements of Drawing 7136036 and this specification; these requirements being detailed herein only to the extent deemed necessary to obtain the desired mechanical and electrical characteristics and performance.

3.3.2 Complete assembly. Headset, Electrical H-154A/AIC shall consist of the following components, assembled in accordance with Drawing 7136036:

Quantity	Item
2 ea.	Earphone, H-143/AIC in accordance with MIL-PRF-25670/2
2 ea.	Shell, Earphone, in accordance with Drawing 7136037
1 ea.	Cord Assembly, Electrical, Branched CX-4707C/AIC in accordance with MIL-DTL-22442/33, P/N: M22442/33-4707.
1 ea.	Cord Assembly, Electrical, Branched CX-4708A/AIC in accordance with MIL-DTL-22442/37, P/N: M22442/37-4708
1 ea.	Installation Kit, Electronic Equipment MK-634/AIC in accordance with Drawing 7136032
2 ea.	Cushion, Earphone Shell, 7136038-10 in accordance with Drawing 7136038
2 ea.	Pad, Earphone, in accordance with Drawing 7136039

3.4 Cord assemblies. Cord assemblies are In accordance with MIL-DTL-22442 specification sheets referenced.

3.5 Performance characteristics.

3.5.1 Electrical.

3.5.1.1 Circuitry. The two earphones shall be connected in parallel, see figure 2 for schematic diagram.

3.5.1.2 DC resistance of the headset circuit. The dc resistance of the headset circuit shall be between 8.5 and 11.5 ohms inclusive (see 4.5.2).

3.5.1.3 DC resistance of microphone circuit. The dc resistance of the microphone circuit shall not exceed the total allowable resistance of the conductors in the circuit (microphone not included in assembly) (see 4.5.3).

3.5.2 Acoustical.

3.5.2.1 Acoustic distortion. There shall be no buzzes, rattles, or voice distortion when the headset is operated as specified herein (see 4.5.4).

3.5.2.2 Functional operation. The headset shall perform satisfactorily when used with Intercommunication Sets AN/AIC-10( ) or AN/AIC-18 and when subjected to the functional operation test specified herein (see 4.5.5).

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T12:06Z
Check the source to verify that this is the current version before use.

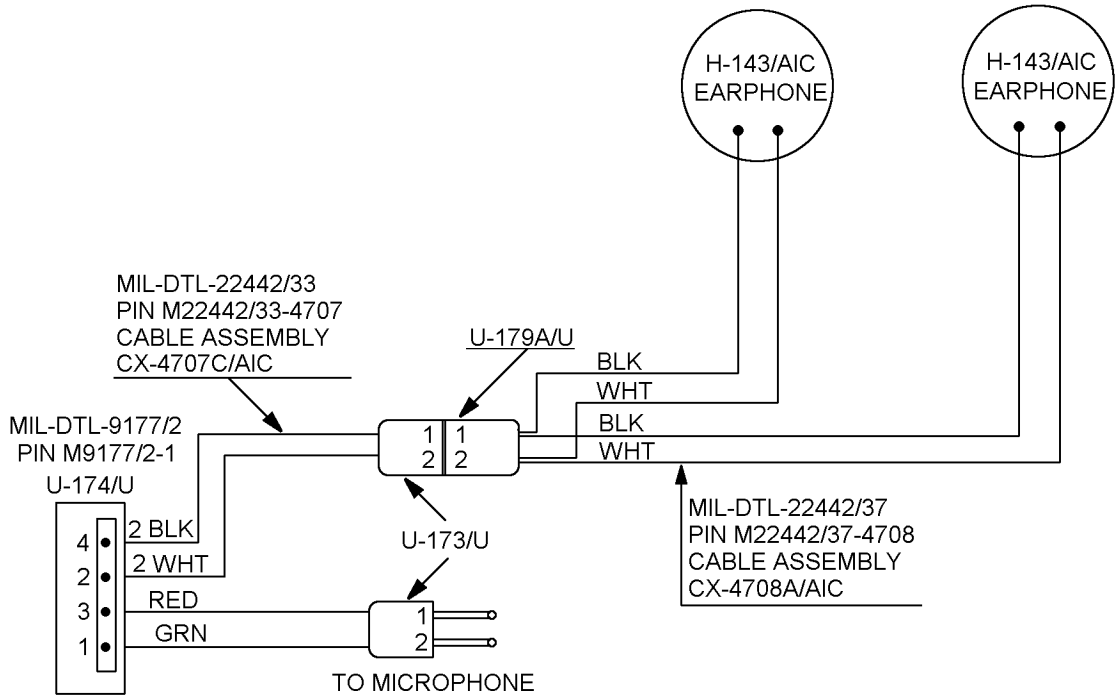


FIGURE 2. H-154A/AIC wiring diagram.

3.6 Environmental tests.

3.6.1 Temperature. When tested there shall be no parts cracked or deformed due to extreme temperature and the DC resistance of the headset circuit requirements of 3.5.1.2 shall be met (see 4.6.1).

3.6.2 Vibration. When tested, there shall be no loose parts or evidence of mechanical failure and the dc resistance of the headset circuit requirements of 3.5.1.2 shall be met (see 4.6.2).

3.6.3 Shock. When tested, there shall be no failure due to broken or deformed parts and the dc resistance of the headset circuit requirements of 3.5.1.2 shall be met (see 4.6.3).

3.6.4 Humidity. When tested, there shall be no failure due to moisture or corrosion and the dc resistance of the headset circuit requirements of 3.5.1.2 shall be met (see 4.6.4).

3.6.5 Salt fog. When tested, there shall be no failure due to corrosion, and the dc resistance of the headset circuit requirements of 3.5.1.2 shall be met (see 4.6.5).

3.7 Marking. Marking shall be in accordance with MIL-STD-1285. Location shall be as specified (see 3.1) and shall consist the following:

a. NSN.

b. PIN.

c. Manufacturer's CAGE and PIN.

3.8 Workmanship. The products shall be processed as to be uniform in quality and shall be free from loose or deposited foreign materials and other defects that will affect life, serviceability, or appearance.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T12:06Z
Check the source to verify that this is the current version before use.

4. VERIFICATION

4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. First article inspection shall be performed on one complete Headset, Electrical H-154A/AIC when first article sample is required (see 3.1). This inspection shall include the examination of 4.4 and the tests of 4.5.1 through 4.5.5.

4.3 Conformance inspection. Conformance inspection shall include the examination of 4.4 and the tests of 4.5.1 through 4.5.4.

4.3.1 First article. A minimum of 10 headsets shall be subjected to the preproduction tests in the quantities indicated in 4.5. At least 3 headsets shall be subjected to each of the preproduction tests. More than one preproduction test may be performed on any one headset provided it has not been damaged by a previous test.

4.4 First article tests. Acceptance tests shall consist of:

a. Individual tests (see 4.4.1).

b. Sampling plan and tests (see 4.4.2).

4.4.1 Individual tests. Each headset shall be subjected to the individual tests listed in 4.5.

4.4.2 Sampling plan and tests. At least one headset shall be selected at random from every 100 headsets produced and be subjected to the environmental tests listed in 4.5.

4.4.2.1 Sampling plan. A sample of parts shall be randomly selected in accordance with table I, if one or more defects are found, the lot shall be re-screened and defects removed. After screening and removal of defects, a new sample of parts shall be randomly selected in accordance with table I, if one or more defects are found in the second sample, the lot shall be rejected and shall not be supplied to this specification.

TABLE I. Sampling plan.

Lot size	Sample size
2 to 8	1/
9 to 15	13 1/
16 to 25	13
26 to 50	13
51 to 90	13
91 to 150	13
151 to 280	20
281 to 500	29
501 to 1200	34
1201 to 3200	42

1/ Indicates entire lot must be inspected or sample size not to exceed lot size.

4.4.2.2 Rejection and retest. When one item selected from a production run fails to meet the specification, no item still on hand or later produced shall be accepted until the extent and cause of failure are determined.

4.4.2.3 Individual tests may continue. For operational reasons, individual tests may be continued pending the investigation of a sampling test failure. But final acceptance of items on hand or produced later shall not be made until it is determined that items meet all the requirements of the specification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T12:06Z
Check the source to verify that this is the current version before use.

4.4.2.4 Sampling plan. Accept on zero, for general inspection as specified in table II.

4.5 Test schedule. The headsets shall be subjected to the tests listed in table II in the quantities shown. The order of performing the first article or acceptance tests is immaterial except that the salt fog test shall be last. At the end of each test, the headset shall be thoroughly inspected and checked for any damage, defects, or deterioration resulting from the tests.

TABLE II. First article and conformance.

Inspection or test	Requirement	Test procedures	Individual	Sample
Examination of product		4.5.1	all	all
DC resistance of headset circuit	3.5.1.2	4.5.2		all
DC resistance of microphone circuit	3.5.1.3	4.5.3		all
Acoustic distortion	3.5.2.1	4.5.4		all
Functional operation	3.5.2.2	4.5.5		all
Temperature-altitude	3.6.1	4.6.1		all
Vibration	3.6.2	4.6.2		
Shock	3.6.3	4.6.3		all
Humidity	3.6.4	4.6.4		1%
Salt fog	3.6.5	4.6.5		

4.5.1 Examination of product. The headset shall be inspected thoroughly to determine conformance to the requirements of this specification with respect to materials, mechanical detail, workmanship, physical dimensions, and finishes, prior to and during assembly. Particular attention shall be paid to neatness and thoroughness of soldering, wiring, marking of parts and assemblies, painting, and screw assemblies.

4.5.2 DC resistance of the headset circuit. The DC resistance of the headset circuit shall be measured across the earphone contacts of Telephone Plug U-174/U (see 3.5.1.2).

4.5.3 DC resistance of microphone circuit. The DC resistance of the microphone circuit shall be measured with the microphone contacts of telephone Plug U-174/U short circuited (see 3.5.1.3).

4.5.4 Acoustic Distortion. Electrical voice signals shall be applied across the earphone contacts of Telephone Plug U-174/U. The input shall be such that the peak of the speech signal applied to the headset is at least 2.0 volts rms. During this test, the operator shall listen for buzzes, rattles, or voice distortion in the acoustic output of the headset (see 3.5.2.1).

4.5.5 Functional operation. A microphone, with impedance between 2.0 to 5.0 ohms, shall be connected to the microphone circuit of the headset, as shown on figure 2. Telephone Plug U-174/U on the headset shall be inserted in a Telephone Jack U-92A/U that has the earphone contacts connected to the output of an amplifier of suitable gain and the microphone contacts to the input of the same amplifier. During this test, the talker shall listen to his own voice while speaking into the microphone to insure that the headset is functioning satisfactorily.

4.6 Environmental testing.

4.6.1 Temperature. The headset or headset-microphone shall be tested in accordance with MIL-STD-810, method 501, procedure 1 with a storage temperature of -55°C and an operating temperature of -40°C and MIL-STD-810, method 502, procedure 1 with a storage temperature of 70°C and an operating temperature of 65°C. Upon completion of the test in 4.5.2 shall be performed (see 3.6.1).

4.6.2 Vibration. The headset or headset-microphone shall be tested in accordance with test method 201 of MIL-STD-202. Upon completion of the test in 4.5.2 shall be performed (see 3.6.2).

4.6.3 Shock. The headset or headset-microphone shall be tested in accordance with test method 213 of MIL-STD-202, test condition A. Upon completion of the test in 4.5.2 shall be performed (see 3.6.3).

4.6.4 Humidity. The headset or headset-microphone shall be tested in accordance with method 103 of MIL-STD-202, test condition B. Upon completion of the test in 4.5.2 shall be performed (see 3.6.4).

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T12:06Z
Check the source to verify that this is the current version before use.

4.6.5 Salt fog. The headset or headset-microphone shall be tested in accordance with test method 101, Salt Atmosphere, of MIL-STD-202, test condition B. Upon completion of the test in 4.5.2 shall be performed (see 3.6.5).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Headset Electrical H-154A/AIC is intended for use with Helmet, Flying HGU-26/P. The headset includes a plug connector to plug in a low impedance dynamic microphone which is mounted in an oxygen mask. The headset is intended for use with Intercommunication Sets AN/AIC-10 and AN/AIC-18 to provide communication of high intelligibility under the extreme noise conditions encountered in military aircraft.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Contract number.

c. When first article is required (see 3.1).

d. Packaging requirements (see 5.1).

6.3 Tin whisker growth. The use of alloys with tin content greater than 97 percent may exhibit tin whisker growth problems after manufacture. Tin whiskers may occur anytime from a day to years after manufacture, and can develop under typical operating conditions on products that use such materials. Conformal coatings applied over top of a whisker-prone surface will not prevent the formation of tin whiskers. Alloys of 3 percent lead have shown to inhibit the growth of tin whiskers (see 3.2.1).

6.4 Environmentally preferable material. Environmentally preferable materials should be used to the maximum extent possible to meet the requirements of this specification. Table III lists the Environmental Protection Agency (EPA) top seventeen hazardous materials targeted for major usage reduction. Use of these materials should be minimized or eliminated unless needed to meet the requirements specified herein (see section 3).

TABLE III. EPA top seventeen hazardous materials.

Benzene	Dichloromethane	Tetrachloroethylene
Cadmium and Compounds	Lead and Compounds	Toluene
Carbon Tetrachloride	Mercury and Compounds	1,1,1 - Trichoroethane
Chloroform	Methyl Ethyl Ketone	Trichloroethylene
Chromium and Compounds	Methyl Isobutyl Ketone	Xylenes
Cyanide and Compounds	Nickel and Compounds	

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T12:06Z
Check the source to verify that this is the current version before use.

6.5 Subject term (key word) listing.

Cord
Cable
Earphone
Microphone
Plug
Tin
Whisker

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

CONCLUDING MATERIAL

Custodians:
Air Force - 11
DLA - CC

Preparing activity:
DLA - CC

(Project 5965-0379-000)

Review activity:
Air Force - 99

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T12:06Z
Check the source to verify that this is the current version before use.